Satellites and How GPS Works

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Introduction to Satellites

A satellite is an object that orbits a larger body in space. Natural satellites include the Moon, which orbits the Earth. Artificial satellites are man-made machines launched by rockets into orbit around the Earth or other bodies. The first artificial satellite was Sputnik 1, launched by the Soviet Union on 4 October 1957. Today thousands of active satellites provide television, internet, weather forecasts, navigation, Earth observation, scientific research and military services. A satellite stays in orbit because it moves sideways at very high speed while falling toward the Earth, so it keeps falling around the planet instead of crashing. Typical low-Earth orbit speed is about 7.8 kilometres per second.

Types of Orbits

Low Earth Orbit (LEO) is at an altitude of about 160 to 2,000 kilometres. The International Space Station, Earth observation satellites and Starlink satellites orbit here, and a satellite takes about 90 minutes to circle the Earth. Medium Earth Orbit (MEO), at roughly 2,000 to 35,000 kilometres, is used by navigation satellites like GPS. Geostationary Orbit (GEO) is at about 35,786 kilometres.

What Is a Geostationary Orbit?

A geostationary orbit is a circular orbit directly above the Earth's equator at an altitude of about 35,786 kilometres. A satellite in this orbit takes exactly 24 hours to orbit the Earth, the same as the Earth's rotation period, and it moves in the same direction as the Earth's rotation. As a result, the satellite stays fixed over the same spot on the Earth's surface. This allows ground antennas, like TV dishes, to point at one location in the sky without tracking. Geostationary satellites are used for television broadcasting, telecommunications and weather monitoring, but because they are so far away, signals take a quarter of a second to make the round trip. The idea was first described by the writer Arthur C. Clarke in 1945. India's INSAT and GSAT series of satellites are placed in geostationary orbit. A related orbit is the geosynchronous orbit, which has the same 24-hour period but may be inclined, so the satellite appears to trace a figure of eight in the sky. Polar and sun-synchronous orbits pass over the poles and are used for Earth observation, because they pass over each place at the same local time every day.

What Is GPS?

The Global Positioning System (GPS) is a satellite navigation system owned by the United States government and operated by the US Space Force. It was developed by the US Department of Defense, with the first satellite launched in 1978, and became fully operational in 1995. GPS is free for civilian use worldwide. A constellation of about 31 operational satellites orbit at an altitude of about 20,200 kilometres, in medium Earth orbit, and each satellite circles the Earth about twice a day. The satellites are arranged so that at least four are visible from almost anywhere on Earth at any time.

How GPS Works

Each GPS satellite carries extremely accurate atomic clocks and continuously broadcasts radio signals containing its position and the exact time the signal was sent. A GPS receiver, such as a smartphone, picks up the signals, measures how long each took to arrive, and multiplies that delay by the speed of light to calculate its distance from each satellite. With the distance from one satellite, the receiver knows it is somewhere on a sphere; with two satellites, on a circle where the two spheres meet; with three satellites, at one of two points, one of which is usually absurd; this method is called trilateration.

The receiver's own clock is not accurate like an atomic clock, so a small timing error would produce a large position error. A fourth satellite signal is used to correct the clock error. So GPS needs a minimum of four satellites to find a position: three to determine latitude, longitude and altitude, and the fourth to correct the timing error. More satellites give better accuracy. GPS also has to apply corrections from Einstein's theories of relativity: satellite clocks run faster because of weaker gravity and slower because of speed, and the net effect, about 38 microseconds per day, would cause errors of kilometres if ignored. A typical smartphone is accurate to about 5 metres outdoors.

Other Navigation Systems and NavIC

Other global systems include Russia's GLONASS, the European Union's Galileo and China's BeiDou. India has developed its own regional system called NavIC, which stands for Navigation with Indian Constellation. It was earlier called the Indian Regional Navigation Satellite System (IRNSS). The first satellite, IRNSS-1A, was launched in July 2013, and the system with seven satellites was completed in 2016, when Prime Minister Narendra Modi named it NavIC. It uses three geostationary and four inclined geosynchronous satellites to cover India and a region extending about 1,500 kilometres around it, with a positioning accuracy better than 20 metres, and it offers a standard service for the public and an encrypted restricted service for the military. Newer second-generation satellites such as NVS-01, launched in May 2023, are being added. India also operates GAGAN, a system jointly developed by ISRO and the Airports Authority of India that improves GPS accuracy for aircraft.

Uses of Satellites and GPS

GPS powers map apps, ride-hailing, food delivery, vehicle tracking, aviation, precision farming, surveying, emergency response and timing for banks and telecom networks. In India, fishermen receive NavIC-based alerts when they approach international boundaries, and railways and public transport use satellite tracking.

Challenges and Future

Challenges include signal blocking in cities and indoors, jamming and spoofing, and space debris. The future includes multi-constellation receivers, satellite internet from low-Earth orbit constellations, and more accurate positioning.

Quick Facts

- GPS needs at least four satellites: three for position and one to correct the receiver's clock.
- A geostationary orbit is about 35,786 km above the equator, with a period of 24 hours, so the satellite appears fixed.
- NavIC (Navigation with Indian Constellation) is India's regional navigation system with seven satellites.
- GPS satellites orbit at about 20,200 km and the system was fully operational in 1995.
- Sputnik 1, launched on 4 October 1957, was the first artificial satellite.